

# Seungeun Lee, PhD

Founder & Principal Consultant · Dr. S Lee Life Sciences

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**A scientist who codes and speaks the domain vocabulary.** Fifteen years across research benches, regulatory bodies, and commercial leadership, with trusted life-science networks across Europe and Korea. Independent practice serving consulting firms, expert networks, and research organizations across EMEA and Asia.

**Two current threads:** **Senior Advisory** (verification and validation, open across engagement shapes and contract forms) and **Agentic AI** (Sense 2026, a life-sciences agentic workspace on active build).

## 01 EXPERIENCE

*May 2026 – Present · Nieuw Vennepe, Netherlands*

### Founder & Principal Consultant

Dr. S Lee Life Sciences

- ◆ **Commercial strategy & partnerships:** Europe–Asia go-to-market and partnership development for biotech. Market sizing, ICP definition, pricing, channel and stakeholder strategy. Sales-operations analysis. Multidisciplinary leadership across sales, project management, lab, and bioinformatics.
- ◆ **Technical & scientific advisory:** R&D support across clinical, environmental, food-safety life-sciences. NGS workflow design (in vivo, in vitro, in silico), method validation, proficiency testing, bioinformatics pipelines (Python, R, Docker). Peer-reviewed manuscripts, technical reports, regulatory documentation.
- ◆ **Digital infrastructure:** Database architecture, LIMS, biobank, sales-operations, reporting; ISO 17025 & 9001 documentation. Platform-flexible: Python, R, SAP, Power BI, SharePoint. Agentic AI workspace design (active build: Sense 2026).

*Mar 2026 – May 2026 · Rotterdam, Netherlands*

### Entrepreneur in Residence

Builders Studio B.V.

- ◆ Domain research, ICP definition, and customer development for an early-stage AI SaaS. Structured interviews with scientific, commercial, and operations leaders across global R&D and biotech, surfacing buyer dynamics and market entry mechanisms.

*Feb 2021 – Feb 2026 · Amsterdam, Netherlands*

### Senior Project Manager · Head of Project Management

MacroGen Europe B.V.

- ◆ **Business development:** EMEA commercialization of scientific innovations; B2B/B2C partnerships, pricing and branding strategies.
- ◆ **Technical engagement:** Advised on workflow trade-offs across in vitro and in silico methods; led proficiency testing, validation reporting, data pipeline design.
- ◆ **Financial planning:** Fiscal planning, revenue forecasting, pricing optimization; reporting infrastructure with SAP, Power BI.
- ◆ **Digital infrastructure:** Organization-wide ISO 17025/9001 systems and LIMS workflows.
- ◆ **Leadership:** Multidisciplinary teams across sales, project management, lab operations, and bioinformatics.

## 02 EDUCATION

*2016*

### PhD, Public Health

Seoul National University

*2013*

### MSc, Molecular Medicine and Biopharmaceutical Science

Seoul National University

*2011*

### BSc, Food and Nutrition

### BSc, Biology

Yonsei University

## 03 SKILLS

- ◆ Commercial strategy
- ◆ Technical & scientific advisory
- ◆ Digital infrastructure
- ◆ Agentic AI & LLM adoption
- ◆ Training & education
- ◆ Cross-functional leadership

## 04 CERTIFICATIONS

- ◆ PRINCE2 7 Practitioner
- ◆ ISO 17025 Awareness & Auditing
- ◆ IATA Dangerous Goods Regulation
- ◆ Big Data Insight
- ◆ Linux for Biologists

## 05 LANGUAGES

- ◆ **Korean:** native
- ◆ **English:** professional
- ◆ **Dutch:** conversational

## 06 PUBLICATIONS

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*Jan 2017 – Jan 2021 · Wageningen, Netherlands*

### **Researcher Genomics · Data Scientist**

Wageningen Food Safety Research (WFSR)

*Stichting Nederlandse Voedsel- en Warenautoriteit (NVWA)*

- ◆ **Data analysis:** Genomic data analysis of food-borne pathogenic microbes using Illumina and Nanopore sequencing, with Python, R, Docker, Ubuntu, Git workflows.
- ◆ **Database management:** Developed biobank database infrastructure using LIMS, RDBMS (Microsoft Access), Jupyter Notebook, Python.

*Apr 2016 – Oct 2016 · Bilthoven, Netherlands*

### **Guest Researcher**

Planbureau voor de Leefomgeving (PBL)

*Netherlands Environmental Assessment Agency*

- ◆ Curated metadata to refine the GLOBIO-Aquatic Projection model. Applied R packages, built regression models, aligned datasets through variable analysis.

*Mar 2013 – Feb 2016 · Seoul, South Korea*

### **PhD Student Researcher**

Environmental Health Engineering Lab

Seoul National University

- ◆ Built a new laboratory from the ground up to optimize in vitro and in silico workflows. Refined NGS biological sequence data with bioinformatics tools to identify environmental pathogens and improve microbial molecular identification.

*Mar 2011 – Feb 2013 · Seoul, South Korea*

### **MSc Student Researcher**

Genome Research Center for Diabetes and Endocrine Disease

Seoul National University Hospital

- ◆ Molecular-based medical analyses: cell culture, differentiation, drug treatments, animal handling, qPCR, bioluminescent assays, immunoblotting, protein extraction, DNA cloning.

*Jul 2009 – Dec 2009 · Davis, California, USA*

### **Research Assistant**

Western Human Nutrition Research Center - USDA, UC Davis

- ◆ Handled essential laboratory equipment, chemicals, and animal models for nutrition and health research applications.

## 06 PUBLICATIONS AND REPORTS

Selected entries only; many similar reports produced routinely.

## Peer-reviewed publications

Indexed via [ORCID 0009-0009-3580-4459](#) with DOI

- 01 Hoek A & [Lee S](#), van den Berg R, Rapallini M, Overbeeke L, Opsteegh M, Bergval I, Wit B, van der Weijden C, van der Giessen J, van der Voort M. (2023). Virulence and anti-microbial resistance of Shiga toxin-producing *Escherichia coli* from goat and sheep dairy farms in the Netherlands. *Journal of Applied Microbiology*.
- 02 [Lee S](#), An C, Xu S, Lee SH, Yamamoto N. (2016). High-throughput sequencing reveals unprecedented diversities of *Aspergillus* species in outdoor air. *Letters in Applied Microbiology*. DOI: 10.1111/lam.12608.
- 03 Xu S, An C, Kim S, [Lee S](#), Lee K, Yamamoto N. (2016). Effects of the biocides on the culturable house dust-borne bacterial compositions and diversities. *Human and Ecological Risk Assessment*. 22(5): 1133-1146.
- 04 [Lee S](#), Yamamoto N. (2015). Accuracy of the high-throughput amplicon sequencing to identify species within the genus *Aspergillus*. *Fungal Biology*. 119(12): 1311-1321.
- 05 [Lee S](#), Xu S, C.P. B, Lee H, Park MS, Lim YW, Yamamoto N. (2015). Triazole susceptibilities in thermotolerant fungal isolates from outdoor air in the Seoul Capital Area in South Korea. *PLoS ONE*. 10(9): e0138725. DOI: 10.1371/journal.pone.0138725.
- 06 [Lee S](#), Kang S, C.P. B, Yoon C, Yang J, Yamamoto N. (2015). Removal of viable airborne fungi from indoor environments by benzalkonium chloride-based aerosol disinfectants. *Human and Ecological Risk Assessment*. 21: 2174-2191.
- 07 [Lee SE](#), Koo YD, Lee JS, Kwak SH, Jung HS, Cho YM, Park YJ, Chung SS, Park KS. (2015). Retinoid X receptor  $\alpha$  overexpression alleviates mitochondrial dysfunction-induced insulin resistance through transcriptional regulation of insulin receptor substrate 1. *Molecules and Cells*. 38(4): 356-361.

## Commercial and market intelligence

Business performance, market analysis, and BI dashboards

- 01 [Lee S](#). (2026). From Data to Decisions: NGS Business Performance Intelligence and 2026 Strategy Blueprint. *Macrogen Europe internal*.
- 02 [Lee S](#). (2026). NGS Master Dashboard: Year Report 2025. *Macrogen Europe internal Power BI dashboard*.

## Inter-laboratory proficiency studies

Proficiency testing and cross-laboratory evaluations

- 01 [Lee S](#), Kook J, Kim J. (2022–2025). Macrogen Europe data submission for: DNA Sequencing - NGS. *European Molecular Genetics Quality Network (EMQN)*.
- 02 [Lee S](#). (2024). Interlaboratory testing at Macrogen: Whole genome, Targeted, and Transcriptome sequencing. *Macrogen Europe internal*.
- 03 [Lee S](#), van der Voort M. (2020). WFSR data submission for: Global Microbial Identifier Proficiency Test 2017. *National Food Institute, Technical University of Denmark (DTU Food)*.
- 04 [Lee S](#), van der Voort M. (2020). Evaluation on the first inter-laboratory exercise on Whole Genome Sequencing of Shiga toxin-producing *Escherichia coli* strains 2017-2018 (PT-WGS1). *WFSR internal validation report for EURL submission*.
- 05 [Lee S](#), van der Voort M. (2020). Evaluation on the 23rd inter-laboratory study (PT23) on the identification and typing of Shiga toxin-producing *E. coli* (STEC) and other pathogenic *E. coli* strains, 2018-2019. *WFSR internal validation report for EURL submission*.
- 06 [Lee S](#), van der Voort M. (2020). Evaluation on the 26th inter-laboratory study (PT26) on the identification and typing of Shiga toxin-producing *E. coli* (STEC), 2019. *WFSR internal validation report for EURL submission*.

## Technical and method validation reports

Method development, validation, and QC workflows

- 01 [Lee S](#), Kook J, Kim J. (2021–2025). NGS Test Method Validation Report: Targeted sequencing. D-NGS-009. *Macrogen Europe internal* (ISO 17025).
- 02 [Lee S](#), Kook J, Kim J. (2025). Internal testing: SureSelect Clinical Research Exome v4. *Macrogen Europe internal*.
- 03 [Lee S](#). (2024). NGS Technical Support Guide, v20240704. *Macrogen Europe internal*.
- 04 [Lee S](#), Park G, Kook J, Kim J. (2023). NGS Low input RNA kit testing: Watchmaker vs. Illumina. *Macrogen Europe internal*.
- 05 [Lee S](#). (2022). NGS Technical Support: PacBio SMRT. *Macrogen Europe internal*.
- 06 [Lee S](#). (2022). NGS Technical Support Guide, v20220315. *Macrogen Europe internal*.
- 07 [Lee S](#). (2021). NGS Technical Support: Single Cell. *Macrogen Europe internal*.
- 08 [Lee S](#). (2021). NGS Technical Support: Metagenome. *Macrogen Europe internal*.
- 09 [Lee S](#). (2021). NGS Technical Support: Sample preparation. *Macrogen Europe internal*.
- 10 [Lee S](#), van der Voort M. (2020). WGS Quality control: BaseClear reads reporting failure in the Per base sequence content. N200100. *WFSR internal*.
- 11 [Lee S](#), Bak A, van der Voort M. (2020). Onderzoeksrapport: Optimization of Nanopore whole genome sequencing. Assembly assessment. R200086. *WFSR internal*.
- 12 [Lee S](#), van der Voort M. (2020). Onderzoeksrapport: Implementing the trim workflow for the quality control of whole genome sequencing reads. R200057. *WFSR internal*.
- 13 [Lee S](#), van der Voort M. (2020). Onderzoeksrapport: Implementing cross-contamination workflow for the quality control of *Escherichia coli* WGS data sets. R200056. *WFSR internal*.
- 14 [Lee S](#), Castelijin G, van der Voort M. (2020). Onderzoeksrapport: Setting cut-off values for the quality control of *Listeria monocytogenes* WGS data sets. R190130. *WFSR internal*.
- 15 [Lee S](#), van der Voort M. (2019). Onderzoeksplan: Optimization of the use of Nanopore whole genome sequencing. N190072. *WFSR internal*.
- 16 [Lee S](#), Rapallini M, Bak A, van der Voort M. (2019). Onderzoeksrapport: Implementation of the Oxford Nanopore Sequencer MinION for Whole Genome Sequencing. R190021. *NVWA internal*.
- 17 [Lee S](#). (2019). Biobank database update manual. *NVWA internal*.
- 18 [Lee S](#), van den Berg R, van der Voort M. (2018). Onderzoeksrapport: Implementatie bioinformatica workflows voor (Shiga Toxin-producing) *Escherichia coli*. R180034. *NVWA internal*.
- 19 [Lee S](#). (2017). LV8.T.Biobanking User Manual. *NVWA internal*.

## Conference presentations

Selected talks - awards where noted

- 01 [Lee S](#). Biobank training: working with scripts. Wageningen Food Safety Research, Wageningen, Netherlands. Jul 30, 2020.
- 02 [Lee S](#). Effects of global temperature change on freshwater fish: temperature sensitivity of freshwater fish. GLOBIO meeting, PBL Netherlands Environmental Assessment Agency, Bilthoven, Netherlands. Aug 30, 2016.
- 03 [Lee S](#), Yamamoto N. Transcriptome analysis of *Aspergillus fumigatus* exposed to triazole fungicides by RNA-Seq. 2nd BK Plus CHEER International Workshop, Seoul National University, Seoul, South Korea. Feb 14, 2016.
- 04 [Lee S](#). Identifying the risks of triazole chemical uses on environmental fungal communities and properties. Graduate Research Competition, Graduate School of Public Health, Seoul National University, Seoul, South Korea. Mar 6, 2015. **AWARDED**
- 05 [Lee S](#), Xu S, C.P. B, Lee H, Park MS, Lim YW, Yamamoto N. Detecting triazole-resistant fungi from ambient air in Seoul, South Korea. 1st BK Plus CHEER International Workshop, Seoul National University, Seoul, South Korea. Feb 2, 2015. **AWARDED**
- 06 [Lee S](#), Kang S, C.P. B, Yoon C, Yang J, Yamamoto N. Removal Effects of Benzalkonium Chloride-based Aerosol Disinfectants on Indoor Airborne Fungi. International Aerosol Conference, Busan, South Korea. Sep 2, 2014.
- 07 [Lee S](#). Current Issue on Aerobiology and Allergy: An Environmental Approach to Public Health. Global Talent Program Workshop, UN Environmental Programme, Bangkok, Thailand. Jun 28, 2013.
- 08 [Lee S](#), Yamamoto N. NGS-based study of fungal allergy. Epidemiological analyses of Environment Seminar, Asan Medical Center, Seoul, South Korea. Jan 8, 2013.
- 09 [Lee S](#), Chung SS, Ahn BY, Koo YD, Park KS. Quantitative recovery of IRS1 with RXR  $\alpha$  activation in mitochondrial dysfunction associated insulin resistance. International Conference on Diabetes and Metabolism, Seoul, South Korea. Nov 10, 2012. **AWARDED**